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A first system/CFD coupled simulation of a complete nuclear reactor transient using CATHARE2 and TRIO_U. Preliminary validation on the Phénix Reactor Natural Circulation Test

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HIGHLIGHTS

- A system/CFD coupling methodology for thermal-hydraulics analysis.
- Application of the model to the Phénix Reactor Natural Circulation Test.
- Validation of the methodology against experimental data.

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ABSTRACT

The natural circulation test (NCT) was conducted in the Phénix prototype French 580 MWth sodium fast reactor (SFR) in 2009. The main goal of the Phénix NCT is to validate system- and CFD-codes with respect to the establishment of natural circulation in the primary system of a pool type SFR. The present paper describes the calculation of the NCT by coupling the 3D computational fluid dynamics (CFD) code TRIO_U with the best estimate thermal hydraulic system code CATHARE. The coupling methodology and the modeling at the system and at the CFD scales are first presented. A validation of the coupling methodology based on a coupled CATHARE/CATHARE calculation compared to the standard CATHARE predictions is then proposed. In a second step, the results of the TRIO_U/CATHARE calculation are compared both to the available experimental data and to the results of a CATHARE alone computation. These comparisons highlight the effectiveness of coupling CFD- and system-codes for the analysis of plant transients where three-dimensional phenomena play an important role.

1. Introduction

System codes such as ATHLET, CATHARE, RELAP, TRAC and TRACE have been widely used these past 30 years to carry a variety of thermal-hydraulic analysis and to support the safety assessment and the licensing procedure for nuclear power plants. These tools have at first been developed for the numerical simulation of postulated accidents in water-cooled reactors such as PWR and BWR. These modeling capabilities have also been used to simulate all kind of experimental facilities, water-cooled research reactors, RBMK reactors, etc. More recently, some of these system codes have been provided with new modeling capabilities including generation IV reactor concepts (Tenchine et al., 2012; Zhou et al., 2012).

In these codes, the mass, momentum and energy balance equations are obtained from a space and/or time averaging of the local instantaneous basic flow equations. The averaging procedure generates a loss of information on local flow processes that must be compensated by an additional modeling effort. Such a work has led to the establishment of a set of dedicated empirical correlations and physical models, the so-called closure laws. The resulting partial differential equations set including the closure laws are generally solved on a mixed 0D–1D grid by means of finite volume/finite difference methods.

However, for some thermal-hydraulic analysis mixed 0D–1D grids are not appropriate. This is the case when dissymmetric physical conditions occur at the entrance of a PWR reactor pressure vessel, e.g. during a boron dilution or a main steam line break scenario. In some other cases, local three dimensional effects in part of a reactor are of highest importance. For instance, thermal

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stratification in the cold leg of a PWR during safety injection is a key issue with respect to aging and plant life extension in a pressurized thermal shock scenario. In pool-type sodium fast reactors (SFR), a large proportion of the primary coolant is comprised in two capacities called the hot- and cold-collectors where three-dimensional effects like thermal stratification or jet mixing take place (Tenchine, 2010).

Some system codes (CATHARE, RELAP5 3D) offer the possibility to locally resolve the flow on a 2D or a 3D grid. However, the obtained flow solutions suffer from low spatial resolution due to the use of coarse meshing. Moreover, since these multidimensional modules have been developed and validated for PWR reactor pressure vessels, their usability for other flow patterns is questionable. On the other hand, computational fluid dynamics (CFD) codes are now able to predict the three-dimensional fluid flow behavior in part of a reactor with reasonable CPU time consumption. Therefore, an increasing attention has being paid these past years to the development of methodologies allowing direct coupling between system and CFD codes both for one- (Bertolotto et al., 2009; Papukchiev et al., 2009; Fanning and Thomas, 2011) and two-phase flows (Aumiller et al., 2001).

The present study describes the system/CFD coupled methodology that has been built recently in the French Atomic Energy and Alternative Energies Commission (CEA) in order to perform one-phase thermal-hydraulic analysis for the French SFR Phénix. This methodology has been developed in anticipation of future demands concerning design and safety issues for the SFR Astrid demonstrator.

The simulating methodology is based on the CATHARE2 system- and the TRIO_U CFD-codes. Both codes include sodium modeling capabilities. The thermal-hydraulic coupling is of the domain-overlapping type associated to a defect correction approach. The software architecture relies on a common application programming interface (API) named ICoCo (Interface for Code Coupling) developed in our group for coupling activities. The numerical methodology was established, previously described and verified on a simple closed loop system configuration (Perdu and Vandroux, 2008).

The objectives of this paper are to present the principal elements of coupling methodology, its application and its validation in a SFR context, to describe the reasoning that have governed the modeling choices and finally to present a comparison between available experimental data and the numerical previsions of the coupled simulation. This numerical/experimental comparison was made against the natural circulation test that took place during the end-of-life Phénix experimental program in year 2009. This comparison is a first illustration of the capability of the methodology applied to a "real" reactor configuration and constitutes a preliminary element of validation of the methodology. This test and the associated modeling strategy are presented in the following section.

The present work is being performed in the framework of the European THINS (Thermal Hydraulics of Innovative Nuclear Systems) project. THINS is devoted to important cross-cutting thermal-hydraulic issues encountered in various innovative nuclear systems such as advanced core thermal-hydraulics, single phase mixed convection, turbulence modeling, multi-phase flows and at last, system-CFD code coupling and validation (THINS web reference).

2. Modeling strategy

2.1. The Phénix reactor and the natural circulation test

Phénix is an industrial/experimental French 580 MWth (initial nominal thermal power) SFR with 35 years of operation. Between the end of electricity production and the start of decommissioning, an end of life test campaign was performed. The NCT took place in 2009 and is part of this research program. The NCT was defined to validate system- and CFD-codes with respect to the establishment of natural circulation regimes in a primary sodium system of a pool type reactor. The Phénix NCT has recently been computed by the CATHARE system code to improve its validation for sodium applications (Pialla et al., 2011). In the frame of an IAEA Common Research Project, several participants from seven member states have also addressed this transient (Monti et al., 2012). In particular, a 3D coarse-grid CFD model of the primary circuit was developed and coupled with an appropriate 1D model for the secondary system (Parthasarathy et al., 2012).

In the Phénix reactor, at nominal state and starting from the diagrid (see Fig. 1), about 90% of the primary sodium flows through the core, the hot collector and the intermediate heat exchangers (IHX). The resulting 10% feed the main vessel cooling system which ends in the cold collector. Consequently, the entire primary mass flow-rate flows through the cold-collector were it is sucked by the primary pumps and forced back into the diagrid. The cold- and hot-collectors as well as the main vessel cooling system are each bounded at their top by a free surface. The primary sodium circuit system is equipped with 3 primary pumps. Initially, each primary pump was surrounded by a pair of IHX belonging to 3 secondary loops. In 1993 a sodium-water reaction permanently damaged the steam generator n° 2. Since then, the maximal thermal power of the core was reduced from 580 MWth to 350 MWth in order to compensate for the loss of a secondary loop. Two IHX have therefore been removed from the primary vessel and the primary pump speed has been reduced. In order to have a better insight into the thermal stratification occurring in the cold-collector, one of the extracted IHX has been replaced by a pole of thermocouples (TC), namely the DOTE TC-pole.

The Phénix NCT takes place at a reduced thermal power of 120 MWth. The complete test is composed of 7 phases but only the 3 first phases are of interest for the present study. Phase 1 is the "unprotected" part of the transient, i.e. it lasts until the occurrence of the reactor scram. It begins by the dry-out of the steam generators which leads, 100 s later, to a temperature rise at the secondary inlet of the IHX. Accordingly, the temperatures in the cold collector, in the primary pumps and at the core inlet increase whereas, due to neutronic feedback effects, the core power decreases. After 458 s, the reactor is manually scrammed and the second phase of the transient begins. At this point, the rotational speed of the secondary pumps is automatically decreased to 110 rpm within 1 min. 8 s after the reactor scram, the 3 primary pumps are manually tripped. Their rotational speed coasts down gradually to 0 thereby enabling the onset of a natural circulation regime in the primary sodium circuit. During this stage, due to the thermal losses of the secondary circuit, approximately 1 MWth are continuously extracted from the primary circuit by the 2 pairs of IHX. This figure is smaller than the total decay-heat of the core which varies between 4 and 2 MWth during this phase. The third phase begins at $t_0 + 10{,}440$ s, by the manual opening of the steam generator casing which initiates a natural circulation of air that constitutes an efficient heat sink for the secondary sodium. This circulation of air is the main contributor to the 3.5 MWth which are continuously extracted from the primary circuit during this phase. The test ends at $t_0 + 24{,}000$ s. Table 1 summarizes these events.

2.2. Specification of the model

Because of its high computational cost, CFD-models must be used only where essential. To decide which part of the Phénix reactor must be modeled with a CFD-code to simulate the natural circulation test, the following reasoning has been made.

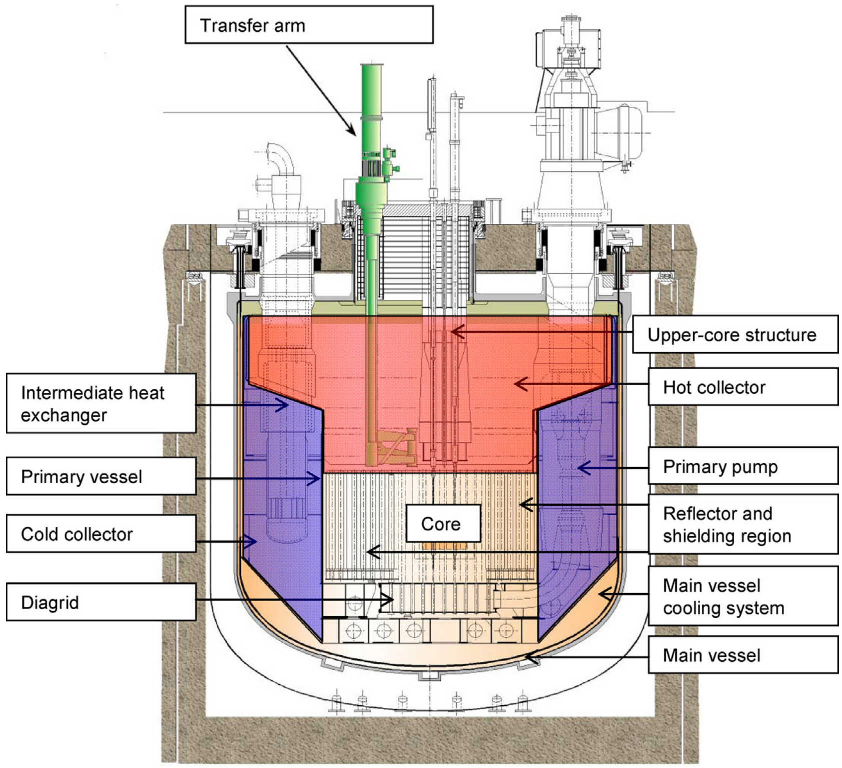

Fig. 1. Schematic of the Phénix primary sodium system.

From the thermal-hydraulic of the primary sodium system point of view, the first phase of the Phénix NCT is characterized by a hot-shock occurring at the IHX primary outlet window. During this unprotected phase of the transient, the temperature at the pump inlet is a key physical parameter for the determination of the core inlet temperature and consequently on the core neutronics. Physically, this parameter is determined by the complex three-dimensional flow paths of the cold-collector. These paths are likely to evolve during the hot-shock because of their sensitivity to buoyancy forces and therefore to the primary IHX outlet temperature. It seems therefore reasonable to prescribe the use of a CFD model for the cold-collector.

After the reactor scram ($t_0 + 458$ s) the second phase begins and natural circulation rapidly takes place in the primary sodium system. A sensitivity analysis performed in our research group has shown that the IHX primary inlet temperature is of greatest importance for the determination of the natural circulation mass flow-rate. This is particularly true in situation where the primary/secondary heat transfer-rate is low. The beginning of the second phase of the Phénix NCT is an example of such a physical situation. Physically, the IHX primary inlet temperature is again determined by 3D complex thermo-hydraulic effects occurring in the hot-collector. The modeling of the hot-collector on CFD bases seems therefore desirable.

Another argument in favor of CFD-modeling for the hot- and cold-collector is related to the instrumentation that produces the experimental data used for the validation. The primary IHX inlet and outlet temperature measurements as well as the core outlet temperature measurements are reliable for the forced convection regime. However, some bias effects, originating from 3D thermal hydraulics phenomena (e.g. thermal stratification, mixing with surrounding sodium, etc.), can be feared in the natural convection

Table 1
Progress of the Phénix NCT along with the main physical phenomena.

Initial state	$t < t_0$	3 primary pumps and 2 secondary loops in operation.	Core power of 120 MWth. Core average entrance and outlet temperatures of respectively 360 °C and 435 °C.
Phase 1	t_0	Water supply to steam generators is manually switched off.	Hot-shock at IHX secondary inlet, temperature increase in the cold-collector, in the primary pumps and at the core inlet. Decrease of fissile power (due to thermal expansion feedback effect).
Phase 2	$t_1 = t_0 + 458$ s	Manual reactor scram. Automatic decrease of the secondary pump rotational speed to 110 rpm within 1 min.	Cold-shock at the core outlet (scram) followed by a hot-shock (primary pump trip). Onset of the natural circulation regime in the primary sodium circuit.
	$t_0 + 466$ s	Manual primary pump trip.	Total heat transfer rate extracted by the IHX of ~1 MWth.
	$t_0 + 4070$ s	Reduction of the secondary pump rotational speed to 100 rpm.	
Phase 3	$t_2 = t_0 + 10{,}440$ s	Opening of the steam generator casing.	Increase of the total heat transfer rate extracted by the IHX to ~3.5 MWth.

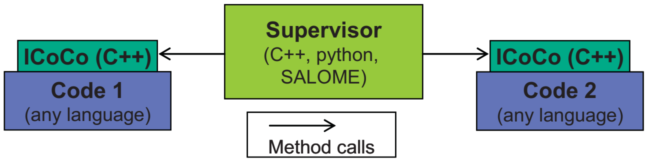

Fig. 2. Schematics of the coupling architecture.

regime. Consequently, for validation purpose, these sensors have to be integrated to CFD-based models able to reproduce the two above-mentioned regimes.

3. Coupling methodology

3.1. Computer architecture

The computer architecture has been designed in order to satisfy a large number of coupling needs, beyond the system-CFD coupled simulation presented here. Hence, a very flexible architecture has been chosen, in order to easily modify the coupling algorithm without modifying the codes involved in the coupling. This architecture is based on the common API ICoCo which is implemented in both codes and will soon be available on the SALOME website (SALOME web reference). A supervisor program then just needs to perform standardized calls on each code through the methods defined in the ICoCo specification (see Fig. 2). The potential loops, conditions and data manipulations are all performed in the supervisor and do not have to be implemented in the codes themselves, resulting in coupling algorithms easier to read, to reuse and to modify.

The supervisor is run on all processors used, and each one opens the relevant code in the form of a dynamic library providing access to an object implementing the ICoCo methods. In the general case, the dynamic libraries do not have to be recompiled when the coupling case changes. Only the codes input decks and the coupling algorithm have to be modified, which is a convenient way to work.

Another benefit of this architecture is that it easily allows the use of parallel interpolators, such as those available with SALOME, from the supervisor. The fields remain distributed on the processors and no bottleneck is introduced.

3.2. General considerations

The available coupling methodologies can be classified into two categories. First, the domain decomposition methods where the system and the CFD domains do not overlap and are coupled to each other through appropriate boundary conditions (BC). Such methodologies are sometimes associated to convergence difficulties stemming from the tight hydraulic coupling between all the junctions separating both domains. Second, the domain overlapping methods where the CFD domain is simulated by both codes and the CFD BC are provided by the system code. At this stage, the method is only one-way and yields zoomed calculations. A possible way to account for a feedback from the CFD computation is to introduce source terms in the system code in charge of minimizing the difference between system and CFD results in the CFD domain. Depending on the physics that is solved by both codes and the confidence that can be placed in the solution found by the CFD code, the user can trigger one, two or all of possible feedbacks namely the mass, momentum and energy feedbacks. This flexibility and a generally good numerical behavior have led our CEA group to the selection of this second method over domain decomposition methods (Perdu and Vandroux, 2008). Fig. 3 shows schematically a reactor and a possible arrangement of the CFD- and system-code domains for a coupled calculation based on a domain overlapping method. In this example, the CFD domain (also called overlapping

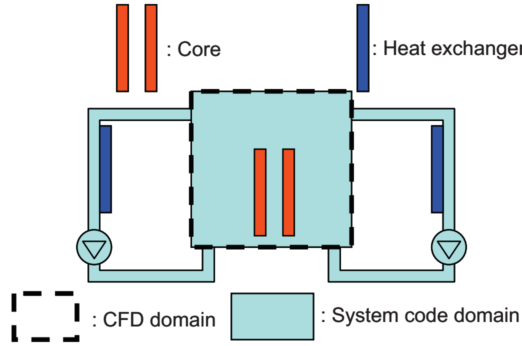

Fig. 3. Schematic of a reactor and typical code domains for a coupled calculation.

domain) is restricted to the core whereas the system code domain include both the core and the loops with their components (pump, heat exchanger, etc.).

The establishment of a domain overlapping method with feedback on energy and/or on momentum applied to a system/CFD coupled calculation was performed recently in our CEA research group (Perdu and Vandroux, 2008). The method was validated on a closed loop configuration comprising a pump, a heat source and a heat sink, for one-phase non-compressible flows (water) and one-phase compressible flows at low Mach number (helium). In the present study, this coupling technique is used for the first time in the context of a pool-type SFR and consequently new features have emerged. The following sections will depict the method in a detailed manner. Its application and validation for pool-type SFR validation will be presented hereafter.

3.3. Boundary conditions for the CFD calculation

A common way to set hydraulic BC to a non-compressible CFD problem consists in imposing mass flow-rates at the inlets and pressures at the outlets. This method was found to produce hydraulic instabilities when used in CFD domains comprising several outlets to an open volume (Perdu and Vandroux, 2008). A more appropriate method is to impose mass flow rates at every hydraulic BC. When the fluid in the CFD domain occupies a fixed volume, the imposed mass flow-rates at the open borders of the CFD domain must be zero sum to comply with the divergence free velocity field calculated by the CFD code. Therefore, when the mass flow-rates are calculated with code which have distinct fluid model properties some small mass-correction are needed. Particular elements relative to the free-surface problem are treated in Section 3.5.

Concerning the thermal BC, when the flow is entering the CFD domain, the enthalpy calculated by the system code is transformed in a temperature and given as a donor cell value to the CFD BC. When the flow is exiting the CFD domain, the same treatment is applied even if the value transmitted by the system code is useless for the CFD calculation. As the BC are similar for inlets and outlets, this method seamlessly allows for flow reversals. In order to be consistent with the physics solved by the system code, the diffusion terms normal to the BC are set to zero.

3.4. Energy and momentum feedbacks

The energy feedback is performed by modifying the energy-balance equation in the system code at every boundary of the CFD domain. These modifications enable to equalize the energy transfers through the boundaries of the CFD domain between the two codes. Since axial thermal diffusion is neglected in the CATHARE system code and because kinetic energy variations can be neglected compared to enthalpy differences in sodium flows, these transfers

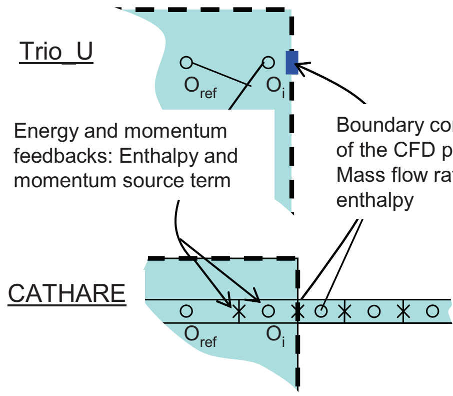

Fig. 4. Coupling scheme between CATHARE and Trio_U: boundary conditions of the CFD problem and energy/momentum feedback to the system-code.

are reduced to the product of the crossing boundary mass flow rate by the convected enthalpy, namely the enthalpy flow-rate.

In the CATHARE energy equation, at every boundary of the overlapping domain, the convected enthalpy is replaced by the enthalpy coming from the CFD calculation. The formulation ensures that all thermo-physical properties (density, viscosity, etc.) of the fluid crossing the boundary are consistent with the CFD value of the enthalpy.

The momentum feedback is designed to match the pressure differences between each boundary and a reference point. Let O_{ref} be a reference point for pressure inside the CFD domain and O_i (i from 1 to N) be the N boundaries of the CFD domain. The target is to obtain $\mathrm{PSYST}(O_{\mathrm{ref}}) - \mathrm{PSYST}(O_i) = \mathrm{PCFD}(O_{\mathrm{ref}}) - \mathrm{PCFD}(O_i)$. One could replace the momentum equation in CATHARE at each boundary by $P(O_i) = \mathrm{PSYST}(O_{\mathrm{ref}}) - \mathrm{PCFD}(O_{\mathrm{ref}}) + \mathrm{PCFD}(O_i)$. This solution was successfully implemented in some simple test cases but is not compatible with the CATHARE code architecture as soon as O_{ref} and O_i are not neighboring points. A more general solution is to implement a correction of the momentum equation through an additional source term regulated to minimize the difference between system and CFD ΔP. This source term is written as follows:

$$\mathrm{DP}(O_i) = \sum_{\text{time step}} \{[P(O_{\mathrm{ref}}) - P(O_i)]_{\mathrm{SYSTEM}} - [P(O_{\mathrm{ref}}) - P(O_i)]_{\mathrm{CFD}}\}$$

$$i = 1 \text{ to } N \qquad (1)$$

From a practical point of view, the right hand sides of the previous equations are updated at each coupling time step by the ICoCo supervisor program by getting the system and CFD pressure values. These correction terms are then set in the system code as additional source terms to the momentum balance equations.

In Fig. 4, a general description of the coupling at the interface between the two codes is shown. However, some additional difficulties must be mentioned. First, the enthalpy of the fluid is not directly available from the TRIO_U code whose principal variable for the energy equation is temperature. Therefore, conversions from temperature to enthalpy and vice versa are required respectively for the energy data coming from or going toward TRIO_U. Second, at the inlets of the CFD domain, the 3D calculation requires information that cannot be provided by the 1D calculation. For instance, a two-dimensional distribution has to be built from the spatially averaged quantities calculated by the system code. The specification of the missing information is a modeling choice that is all the easier to do than the flow is simple at the 1D/3D interface. Consequently, the positioning of the 1D/3D interfaces must be done carefully.

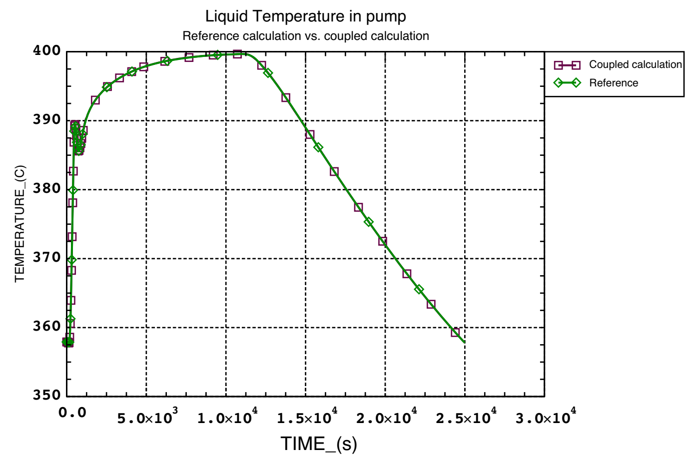

Fig. 5. Comparison between the CATHARE/CATHARE coupled calculation and the reference CATHARE calculation of the Phénix NCT.

3.5. Dealing with the free surface

The two domains that have been chosen to be simulated by the CFD code TRIO_U are shown in Fig. 6. They correspond respectively to the primary cold- and hot-collectors. Thus the TRIO_U CFD models of the cold- and hot-collectors are bounded at their top by a horizontal open BC with an uniformly imposed mass flow-rate. When sodium is entering the CFD domain by this BC, a realistic external temperature has to be prescribed to the model. Therefore, the CFD models of both collectors are each associated to a simplified 0D model which solves the energy balance of the fluid above the BC in order to evaluate its temperature. The energy balance includes the contributions of the enthalpy flow-rates through the horizontal BC and the heat losses across the argon layer toward the concrete roof. This last phenomenon is accounted for with the same empirical correlation used in the CATHARE stand-alone simulation.

Setting a correct external pressure at the horizontal BC is necessary to account for a proper momentum feedback from the CFD model to the system model. Considering that sodium above the horizontal BC lays in hydrostatic equilibrium allows for directly determining the external pressure from its mass. Therefore, the CFD models of both collectors are associated to a 0D model that solves the mass balance equation for the sodium above the horizontal BC. At this stage, it is worth recalling that non-expandable fluid properties are used for the CFD calculations. Therefore, the mass flow-rate

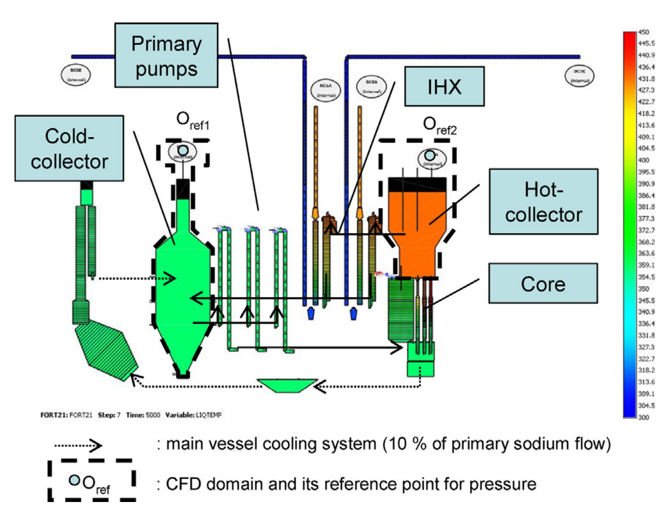

Fig. 6. CATHARE nodalization of the Phénix reactor, the CFD domains and the calculated thermal field at 120 MWth.

through the horizontal BC computed by the CFD model does not account for thermal expansion in the cold- and hot-collectors. Since the geometry of the cold- and hot-collectors is non-cylindrical, this assumption is a source of inaccuracy regarding free surface level variations and thus pressure in the CFD domain. To overcome this problem, an additional thermal expansion mass flow-rate through the horizontal BC was introduced in the mass balance equation of the 0D models. This thermal expansion contribution to the mass flowrate is calculated as $\int_{\Omega} \beta\rho(\partial T/\partial t)d\Omega$.

The derivation of this thermal expansion contribution is performed as follows. Let Ω and $\delta\Omega$ be respectively the CFD computational domain and its boundary. The mass balance equation over Ω for a non-expandable, non-compressible fluid reads:

$$\int_{\partial\Omega} \rho\vec{u}.\vec{n}dS = 0 \tag{2}$$

where ρ, u and n respectively denote the density, the velocity and the normal direction on $\delta\Omega$ oriented outward. The same equation but for an expandable fluid is:

$$\frac{d}{dt}\int_{\Omega} \rho_{ex} d\Omega + \int_{\partial\Omega} \rho_{ex}\vec{u}_{ex}.\vec{n}dS = 0 \tag{3}$$

where the "*ex*" subscripts refers to physical quantities obtained from a CFD simulation performed with expandable fluid properties. Considering that the mass flow-rates are the same for both CFD models through every open boundary except the horizontal BC, the difference between the latter equation and the previous one gives the thermal expansion mass-flow rate that is looked for:

$$\int_{\text{horizontal BC}} \rho_{ex}\vec{u}_{ex}.\vec{n}dS - \int_{\text{horizontal BC}} \rho\vec{u}.\vec{n}dS = -\frac{d}{dt}\int_{\Omega} \rho_{ex} d\Omega \tag{4}$$

The exact evaluation of the right hand side of the previous equation requires the temperature field from a CFD simulation performed with expandable fluid properties. In the present work, such a simulation was not done. Instead, the following approximate estimation, relying on the solution obtained from the non-expandable CFD simulation, is proposed:

$$-\frac{d}{dt}\int_{\Omega} \rho_{ex} d\Omega \approx \int_{\Omega} \beta\rho\frac{\partial T}{\partial t} d\Omega \tag{5}$$

where β and T respectively stand for the thermal expansion coefficient and temperature.

3.6. Validation of the coupling methodology

In view of validating the system/CFD coupled calculation with respect to experimental data, it is useful to validate the coupling methodology first. In this section, a validation of the coupling methodology is proposed. The validation relies on the verification that the results of a reference simulation can be reproduced by a coupled calculation. To ease the reference/coupled comparison the CATHARE code was used in both computations, for the CFD domains as well as for the system domain.

The validation is performed for the Phénix NCT relying on the CATHARE input deck and with the CFD domains depicted in Fig. 6. The CATHARE/CATHARE coupled calculation and the CATHARE reference calculation are launched with a unique ICoCo supervisor program in order to impose a common time step to all processes. An explicit time-scheme described was selected to perform the CATHARE/TRIO_U coupled calculation.

Fig. 5 shows a comparison between the CATHARE/CATHARE coupled calculation and a reference CATHARE stand-alone calculation for the evolution of the sodium temperature at pump entrance during the Phénix NCT. As can be seen in Fig. 5 quasi-perfect agreement is found between the two calculations. For instance, temperature errors lower than 0.1 °C were recorded for the whole transient using a time step of 2 s.

The results of this verification test case show that the chosen coupling methodology produces quasi-exactly the same answer as a reference stand-alone calculation.

4. Development of the system/CFD coupled model

4.1. System scale model

The system-scale model used for the coupled calculation derives from the data input deck prepared in our research group to compute the Phénix NCT with the CATHARE code (Pialla et al., 2011). This system-scale model represents completely the thermal-hydraulics of the primary sodium system, i.e. it encompasses the primary pumps, the core, the IHX, the cold- and hot-collectors including the upper-core structure (UCS) and the main vessel cooling system. The sweeping of the free surface by argon gas is modeled by three 0D modules located at the top of respectively the cold-collector, the hot-collector and the main vessel cooling system. All three are connected to an open boundary condition with a fixed external pressure. The other general features of the model are the following. The heat losses toward the roof and the emergency cooling circuit are accounted for by specific empirical correlations. In order to focus on the thermal-hydraulics, the neutronic and residual powers are provided to the model by means of an external user-law. To facilitate the comparison between numerical and experimental plant data, the events occurring during the transient are triggered in the computation according to a time criteria instead of a physical criteria.

An advantage of domain overlapping methods is that coupling can supposedly be performed without modifying the model at the system scale. Nevertheless, some minor changes had to be made in the CATHARE data input deck. First, the meshing of the vessel cooling system has been refined in order to ease its thermal coupling with the cold-collector (CFD domain). Second, the modeling of the cold- and hot-collectors was simplified as possible. This simplification has no impact on the solution of the coupled model since it is performed inside the overlapping domain where the results of the system code are anyhow corrected by those of the CFD calculation. Third, since the system code and CFD models were built independently, some geometrical discrepancies have appeared between the two models. This led us to slightly modify some geometrical values (elevations, cross-section, etc.) in the system-scale model. The CATHARE nodalization obtained after this preparation work is depicted in Fig. 6.

4.2. CFD model of the hot- and cold-collectors

4.2.1. Mesh, turbulence model, BC

Firstly, the CFD domains represent a total volume of several hundreds cubic meter. Secondly, walls are of greatest importance for the phenomena at stake because of their role in defining hydraulic paths and hot-/cold-collectors heat exchanges. Thirdly, the required mesh size to simulate the initial steady state is several orders of magnitude smaller than that of the CFD domain. Consequently, only a coarse CFD grid can be considered for the present problem. Undoubtedly, this meshing will not meet the requirements specified by the users' guidelines generally followed by the nuclear community (CSNI, 2007). Therefore the validation procedure performed against the experimental data available for the Phénix NCT is essential.

It is also worth mentioning that very sudden changes in thermal- and hydraulic flow regimes are likely to occur in the CFD domains during the NCT. Therefore a stable upwind convection scheme and

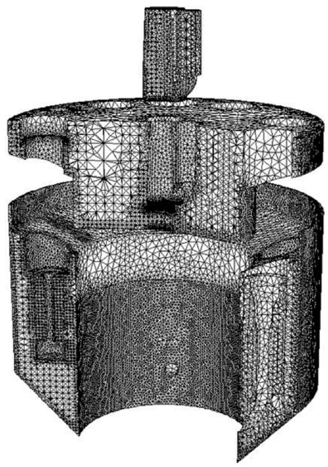

Fig. 7. Grid used for the CFD calculations of the cold-collector, the hot-collector and the upper-core structure.

a second order implicit diffusion scheme have to be selected. The calculation of the Phénix NCT requires the simulation of 3200 s for the establishment of the initial steady state and 25,000 s for the transient. Consequently, an implicit Euler time scheme allowing the use of large time steps is needed, and time consuming models must be avoided. Thus, the Large Eddy Simulation approach and two-phase models (for the free surface) were not considered in the present work. For the same reason, a Boussinesq non-compressible model was selected instead of a quasi-compressible (i.e. dilatable) model.

The preceding considerations have led us to the design of the calculation grids depicted in Fig. 7 comprising respectively 880,000, 350,000 and 50,000 meshes for the cold-collector, the hot-collector and the UCS. The CFD domain of the cold-collector expands into both the IHX and the primary pumps in order to locate the 1D/3D interfaces in regions where the flow is close to one-dimensional conditions. For the same reason, the CFD domain of the hot-collector penetrates the IHX toward the primary inlet window.

The Phénix NCT is characterized by low to moderate Reynolds number flows (up to a few 10^4), during phases 2 and 3. In order to limit computational effort, it was decided not to rely on a k-ε type turbulence modeling. A sub-grid turbulence model of the Smagorinsky type (with Cs = 0.1) was tested for the hot-collector but such modeling did not result in any significant improvement nor modifications of the numerical results (see Section 5.4). Thus, no turbulence model was used for the subsequent calculations.

As already pointed out in the section devoted to the coupling methodology, it was decided in this work to impose the mass flow-rates calculated by CATHARE at every 1D/3D interface. For all these BC, the imposed mass flow-rates was scattered into a two-dimensional uniform velocity and temperature distribution.

4.2.2. Numerical performances

The simulation of the Phénix NCT was performed using time steps of the order of 0.3 s during phase 1 and 1 s after. The stability timestep, which is defined as $dt_{stab} = [(dt_{conv})^{-1} + (dt_{diff})^{-1}]^{-1}$, where the convective time step $dt_{conv} = (|Ux|/\Delta x + |Uy|/\Delta y + |Uz|/\Delta z)^{-1}$ follows Courant Friedrich Levy Criterium and the diffusive time step $dt_{diff} = (2\nu)^{-1}(1/\Delta x^2 + 1/\Delta y^2 + 1/\Delta z^2)^{-1}$ follows the Fourier criterion, is equal to 0.002 s in phase 1 and 0.004 s after. Due to the use of an implicit time marching methodology (of implicit Euler type), the time step used can reach one hundred of stability time steps.

A total of 30 CPU were used to run the model leading to a calculation time of approximately 30 s per time step representing a total of 8 days for the simulation of the steady state and the NCT.

4.2.3. Thermo-physical properties

The thermo-physical properties of sodium used for both CFD problems of the collectors are considered constant and are listed in Table 2. The relative differences in the density functions used in CATHARE and TRIO_U for the evaluation of the buoyancy forces where lower than 5.7×10^{-5} in the [320 °C; 480 °C] temperature range.

4.2.4. The upper-core structure

In the present work, the role of the UCS during NCT is addressed using a pure solid thermal CFD problem thermally coupled to the CFD problem of the hot-collector. Density and heat capacity where chosen so that the global thermal inertia of the CFD model corresponds to that of the actual UCS. The thermal conductivity of the UCS was chosen equal to that of sodium in the reference coupled calculation. This assumption was supplemented by a sensitivity calculation (see Section 5.4).

4.3. Thermal coupling

Heat transfers through the primary vessel (see Fig. 1) and through the hydraulic wrappers (steel layer separating the main vessel cooling system from the cold-collector) are very important to consider for obtaining a correct prediction of the cold-collector thermal stratification. Thus, a thermal coupling routine was written in the supervisor program to address this issue. The routine allows exchanging temperatures and heat fluxes between two CFD problems or one CFD problem and the CATHARE system code model. To cope with non-conformal meshes and parallelism, a SALOME toolbox dedicated to field interpolation and projection was used.

For the hydraulic wrappers, the heat transfer-rate was inferred from a simplified model considering three thermal resistances in series respectively corresponding to sodium convection, conduction in the wall and sodium convection again. For the primary vessel which separates the cold- and the hot-collectors, a purely conductive thermal resistance based on an equivalent steel thickness was considered. This is justified by the fact that the conductive resistance was found to be predominant both in the forced and the natural circulation regimes. Moreover, due to some geometrical details of the primary vessel that could not be implemented in the model, a reduced heat exchange surface is considered by the model. The equivalent steel thickness was tuned in order to obtain a correct initial thermal stratification in the cold-collector. A sensitivity calculation has been performed to check that an increase of the equivalent steel thickness during phases 2 and 3 did not alter significantly the results.

5. Comparison between numerical and experimental results

5.1. Experimental data

The numerical vs. experimental comparison has been performed against measured quantities consisting of inlet/outlet primary and secondary IHX temperatures, primary pump inlet temperature, core outlet temperature, primary sodium mass flow-rate as well as level measurements in the hot-collector, cold-collector and in the main vessel cooling system. The experimental data are the same as those used in the context of the dedicated coordinated research program set by AIEA on the Phénix NCT (Monti et al., 2012). Vertical temperature profiles in the cold- and hot-collector have also

Table 2
Sodium thermo-physical properties used for the CFD calculations.

Density ($kg\,m^{-3}$)	Reference temperature (°C)	Thermal exp. coefficient (K^{-1})	Thermal conductivity ($W\,m^{-1}\,K^{-1}$)	Thermal capacity ($J\,kg^{-1}\,K^{-1}$)	Dynamical viscosity ($kg\,m^{-1}\,s^{-1}$)
856.48	400	2.75×10^{-4}	66	1280.7	2.45×10^{-4}

been recorded by four different poles of TC (see Fig. 8), namely EIM, DOTE, COLTEMP and ML03, during the Phénix NCT but these data cannot be published in the open literature yet.

5.2. Initial 120 MWth steady state

Obtaining the 120 MWth steady state requires a several thousands seconds CFD/system coupled calculation to correctly evaluate the cold-collector thermal stratification. Stratification originates from both the heating by the hot-collector and the cooling by the main vessel cooling system. It occurs at elevations where sodium is poorly mixed by the primary flow, i.e. above the top of the IHX outlet windows. Thermal stratification of the cold-collector was found to significantly alter the sodium temperature at pump inlet during phase 1 of the NCT.

The primary mass flow-rate, as well as the temperature at the inlet/outlet of the main components provided by the CFD/system coupled calculation where found to be very similar to that of the system stand alone calculation for the initial steady state. The only significant difference resides in the main vessel cooling mass flow-rate that was reduced in the coupled calculation in order to obtain a better prediction of the initial cold-collector thermal-stratification.

5.3. The NCT transient

Figs. 9, 11, 12, 14, 15, 17, 18 presented in the subsequent paragraphs all obey to the following rules. The experimental data are plotted with green dotted lines. The numerical results obtained from the CFD/system coupled calculation and from the system alone calculation are respectively plotted with blue and red lines. The results of the system alone calculation are those described in (Pialla et al., 2011). The beginning of the transient is arbitrary set at $t_0 = 5000$ s.

5.3.1. IHX primary outlet temperature

The transient evolution of the primary IHX outlet temperature, as measured by the TC belonging to the EIM pole at an elevation of −4200, is shown in Fig. 9. The EIM-4200 TC is located at the bottom of the IHX outlet window and is immersed in the sodium stream exiting the IHX in the forced convection regime. The 60 K increase resulting from the steam generator dry out during phase 1 and during the beginning of phase 2 is very well reproduced by the CFD/system coupled calculation. A 15 K temperature drop is then observed at $t = 5490$ s, 24 s after the primary pump trip. Visualization of 3D temperature and velocity fields have revealed (see Fig. 10) that this temperature drop originates from the transition

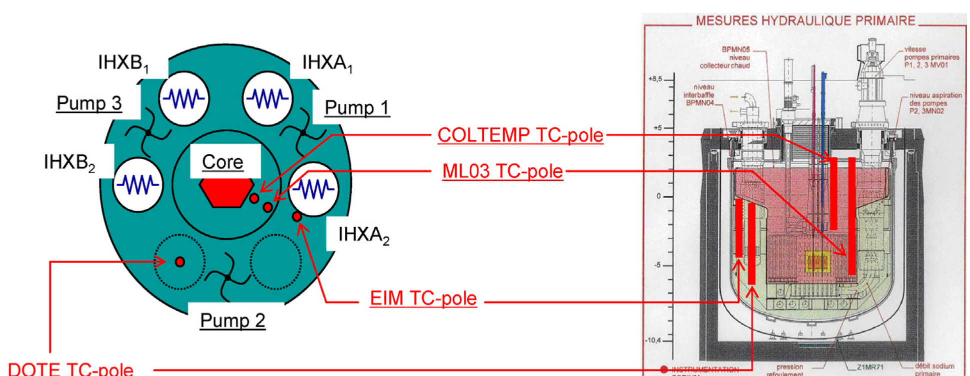

Fig. 8. Horizontal (left) and vertical (right) cross-sectionals views of the primary sodium system and the TC poles.

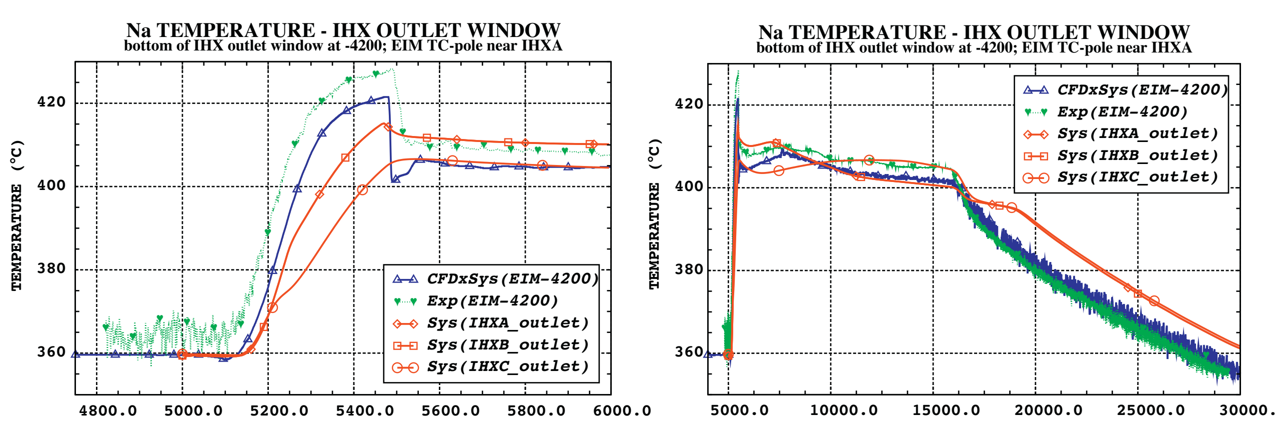

Fig. 9. Short- (left) and long- (right) term evolutions of the IHX primary outlet temperature.

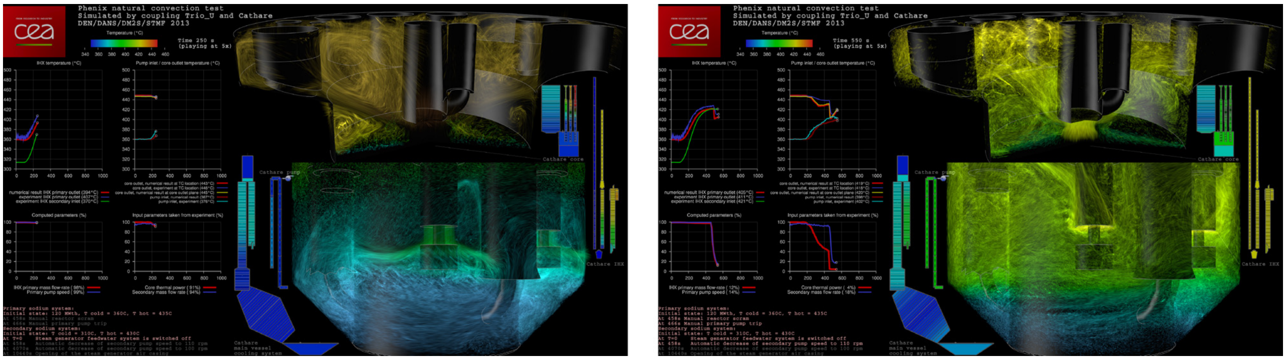

Fig. 10. Temperature field and primary sodium flow seeded by particles during phase 1 at $t_0 + 250$ s (left) and after primary pump trip at $t_1 + 92$ s (right) according to the CATHARE/TRIO_U coupled calculation.

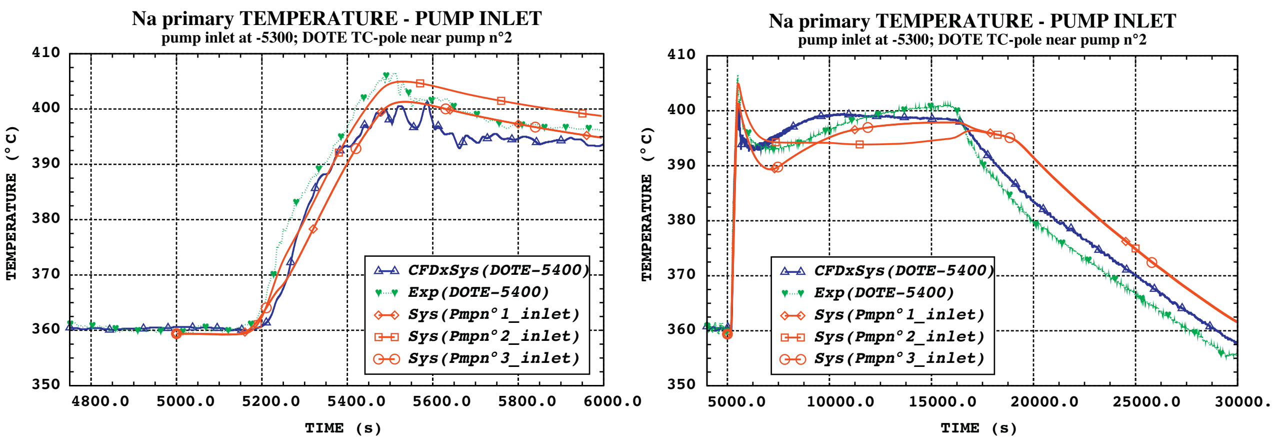

Fig. 11. Short- (left) and long- (right) term evolutions of the primary pump inlet temperature.

between the forced convection to the natural circulation regimes. Indeed, when buoyancy forces become predominant, the sodium stream exits the IHX at the top of the outlet window and the measurements of EIM-4200 TC experiences a sudden shift from the average outlet temperature of the IHX to the local temperature of the cold-collector which happens to be 15 K lower than the former one at this stage. The main streamlines of the flow shown in Fig. 10 were made visible by seeding the flow with virtual particles and setting an appropriate exposure time, so that each particle appears as a streak in the picture. This post-processing technique and the corresponding software named Snorky3D has been developed and described in (Mathieu, 2012). Video 1 (only for the electronic version of the article) presents a visualization of the primary sodium flow in both collectors using snorky3D during phase 1 and the beginning of phase 2.

The predictions of the CFD/system coupled calculation are also in very good agreement with the measured values during phases 2 and 3.

To illustrate the effectiveness of coupling CFD and system codes in addressing SFR transient analysis, the predictions of the system code alone (Pialla et al., 2011) are also shown in Fig. 9. Three liquid temperatures corresponding to the 0D elements located at the

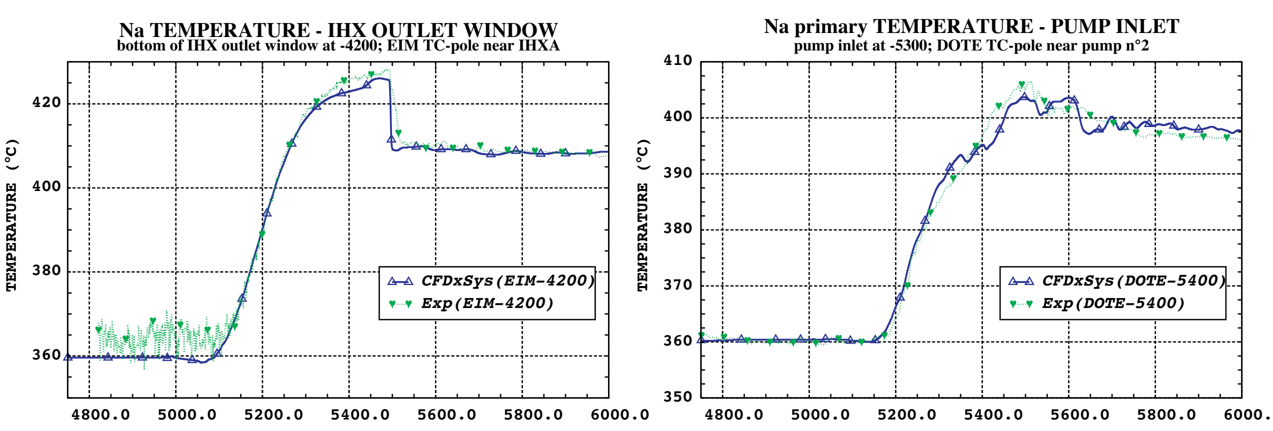

Fig. 12. Short-term evolutions of the primary IHX outlet temperature (left) and of the primary pump inlet temperature (right) obtained with modified BC at the secondary IHX inlet.

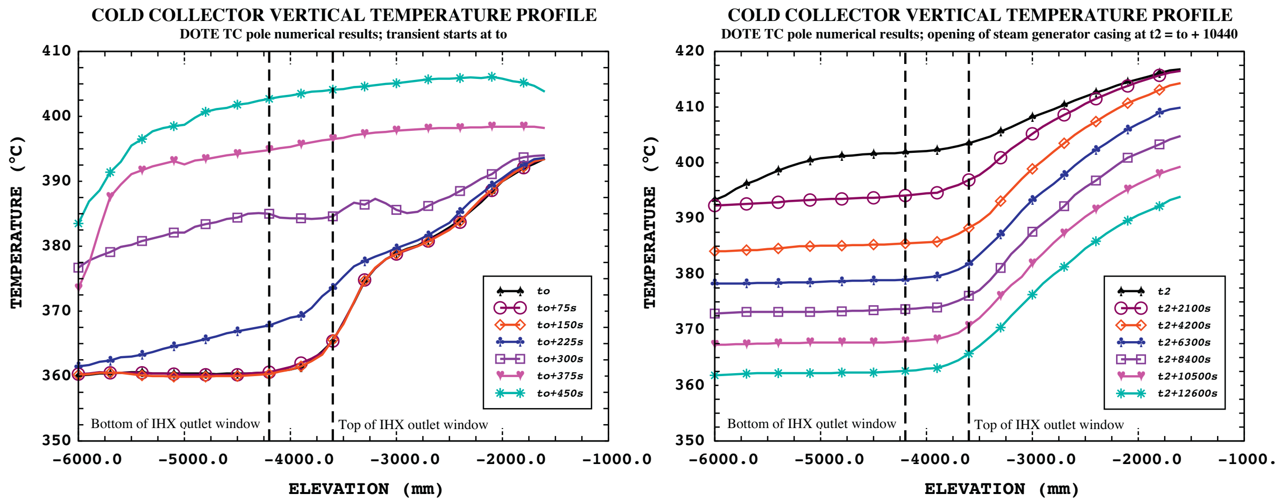

Fig. 13. Phase 1 (left) and phase 3 (right) cold-collector vertical temperature profiles: numerical results.

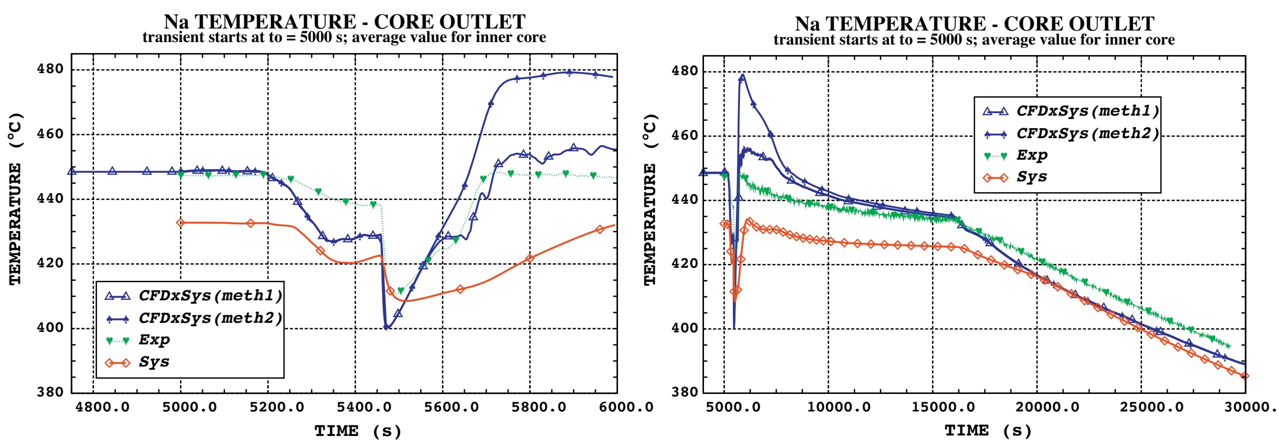

Fig. 14. Short- (left) and long- (right) term evolutions of the mean inner-core outlet temperature.

elevation of the EIM-4200 TC are presented since the system code model of the cold-collector is azimuthally sectorized into three equal parts. The reader interested in the description of the system code model can refer to (Pialla et al., 2011). Obviously, the local 3D phenomena leading to the experimental evolution around $t_0 + 490$ s are not resolved by the system code. More generally, the spatially averaged quantities calculated by the system code present smoother evolutions than the experimental values obtained from local measurements. If correct, the temperature field obtained from the CFD/system coupled calculation may eventually be used to define the spatially averaged quantities that could be directly compared to the results of a system code. This could lead to a more accurate validation procedure for the use of system codes to simulate SFR.

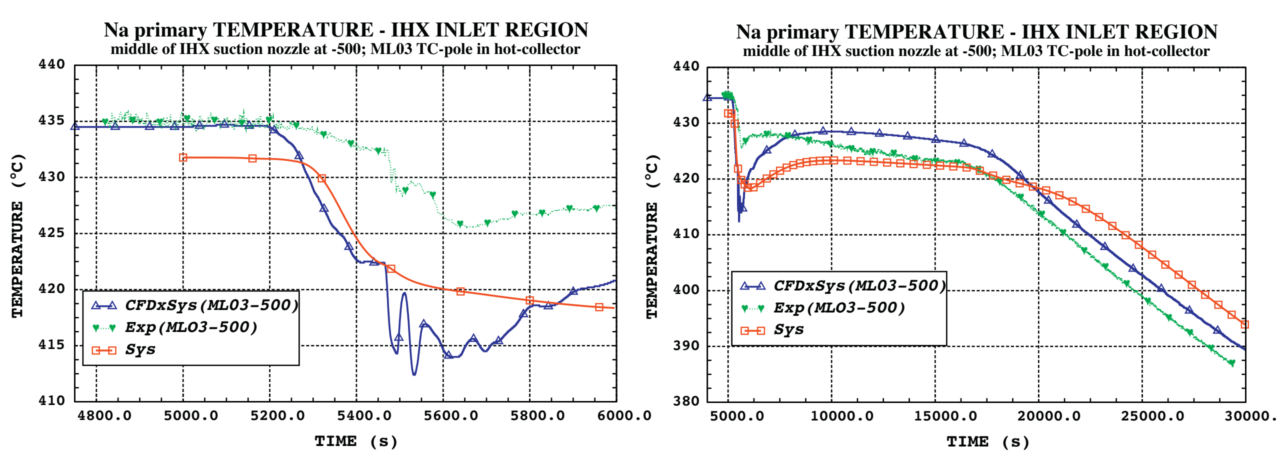

Fig. 15. Short- (left) and long- (right) term evolutions of the primary IHX inlet temperature.

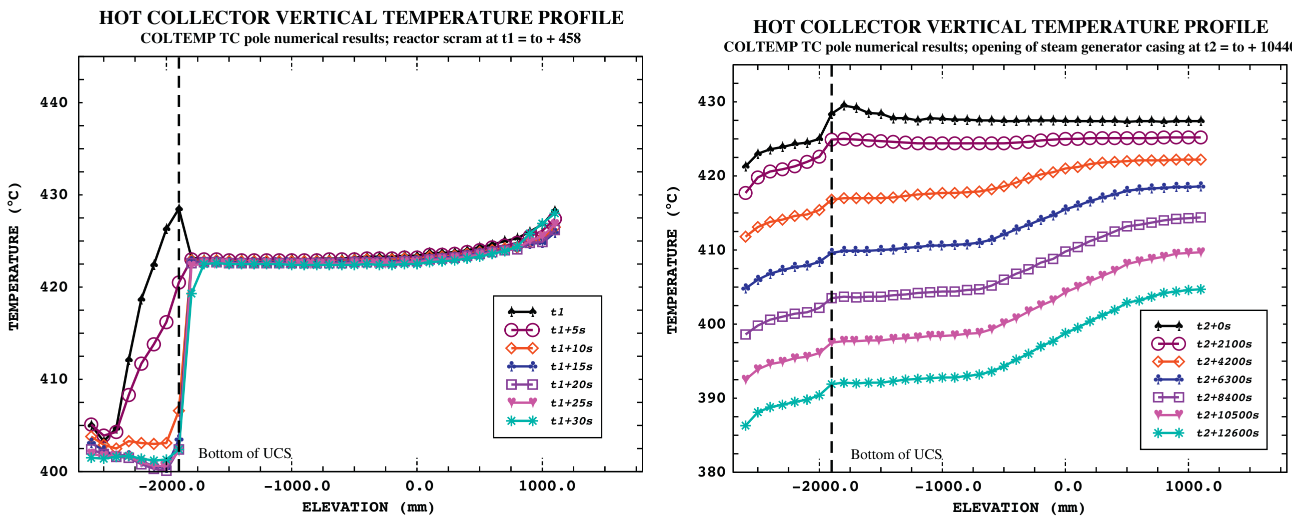

Fig. 16. Hot-collector vertical temperature profiles after reactor scram (left) and during phase 3 (right): numerical results.

5.3.2. *Primary pump inlet temperature*

Fig. 11 shows the transient evolution of the pump inlet temperature, as measured by the TC belonging to the DOTE pole at an elevation of -5400. The experimental data and the numerical predictions of the CFD/system calculation agree reasonably well. However, some significant discrepancies can be observed and have to be commented.

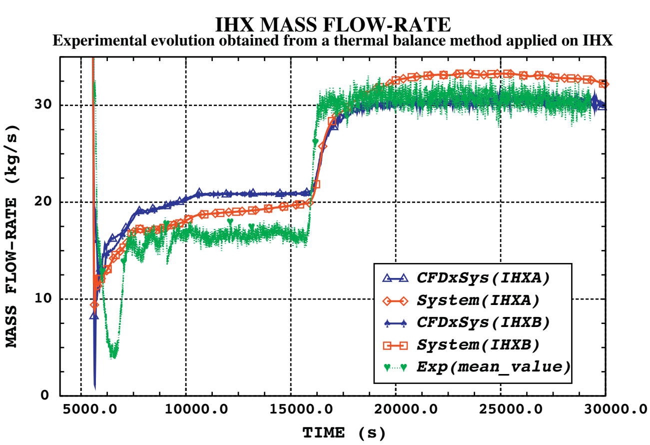

Fig. 17. Experimental and numerical evolutions of the IHX primary mass flow-rate.

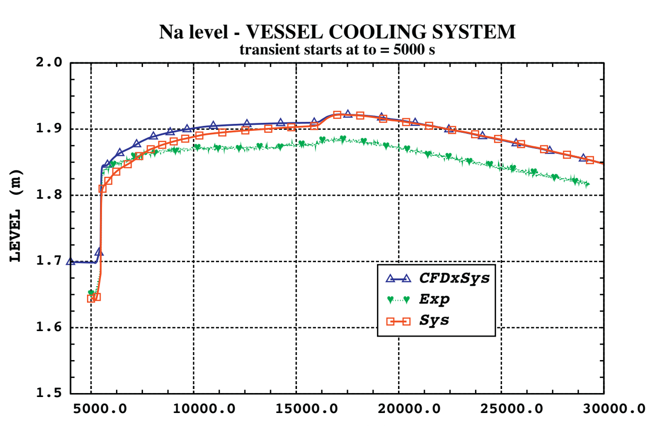

Fig. 18. Experimental and numerical evolutions of the vessel cooling system level.

First, the coupled calculation systematically under-estimates the temperature at the pump entrance during the first 1000 s of the NCT. This evolution depends on the initial thermal stratification of the cold-collector. Indeed, the more stratified the cold-collector, the hotter the sodium at the pump entrance during phase 1. However, several additional calculations performed with a modified initial thermal stratification of the cold-collector have not been able to significantly reduce the gap between experimental and numerical results. A more convincing explanation relies on the fact that the measurements that have served to establish the temperature BC at the secondary IHX inlet are inaccurate when the temperature rate of change is high. The time constant of these TC in their housing can be estimated to be greater than 30 s and the temperature rate of change peaks at +0.7 °C/s during phase 1. These conditions can lead to significant temperature errors. The results of a sensitivity calculation performed with a 40 s upstream translation in time of the BC at IHX secondary inlet supports this idea. Fig. 12 shows that the experimental vs. numerical agreement is greatly improved in this sensitivity calculation both for the primary IHX outlet temperature and the primary pump inlet temperature.

Another interesting feature visible on the right part of Fig. 11 is that the experimental dynamics of the pump inlet temperature is not well reproduced by the coupled numerical model during phase 2. This may originate from the fact the heat exchanges between the cold-collector and the reflector and shielding region of the core (see Fig. 1) are neglected in the present model. The realization of an improved model including these exchanges is underway to test this assumption.

5.3.3. *Cold-collector vertical temperature profile*

The cold-collector vertical temperature profile is analyzed along the DOTE TC-pole. This pole is supported by a plug positioned at the location of one of the missing IHX (see Fig. 8). In this paragraph, only the numerical predictions of the CATHARE/TRIO_U coupled calculation are presented.

The left part of Fig. 13 presents snapshots of the cold-collector vertical temperature profile during phase 1. At the initial state, the cold-collector is thermally stratified above the IHX outlet window. The vertical temperature gradient variations are maximal at the top of the IHX window ($z = -3600$) and at the inlet of the main vessel cooling system ($z = -2550$). An average vertical gradient of 35 °C/5 m is predicted by the coupled calculation. On the contrary, the part of the cold-collector under the IHX outlet window is isothermal. During phase 1, the bottom of the cold-collector is first affected by the hot shock. Later, the hot-shock also modifies the temperature field in the top of the cold-collector. At the end of phase 1, the top of the cold-collector is isothermal and well mixed by the primary sodium flow. A transition from stratified to mixing conditions is experienced by the cold-collector during this phase. This phenomenology can also be observed in Video 1.

The right part of Fig. 13 presents the same snapshots as previously but focuses on phase 3. The bottom of the cold-collector is thermally stratified at t_2 whereas it is isothermal and well mixed by the primary flow from $t_2 + 2100$ s to the end of phase 3. Again, this transition from stratified to mixing conditions occurring at the beginning of phase 3 can also be seen in Video 2.

5.3.4. Core outlet temperature

Each fuel sub-assembly outlet temperature is monitored by a TC suspended to the upper core structure and located 10 cm above the core exit plane. Fig. 14 shows the experimental evolution of the mean value obtained from the TC above the inner-core fuel sub-assemblies. During phase 1, this parameter diminishes because of the hot-shock at the core inlet which causes a decrease of the core fissile power. In the coupled and system alone calculations, the fissile power is fixed by a user external law that has been established using in core measurements. The same remark also applies to the decay heat power after the reactor scram except that the external law relies on the results of a third calculation performed with a CEA dedicated code named DARWIN. Thus, the under-prediction by the model of the primary pump inlet temperature (see Fig. 11) mechanically leads to an under-prediction of the core inlet temperature and consequently to an under-prediction of the mean core outlet temperature during phase 1.

At the beginning of phase 2, the reactor scram followed 8 s later by the primary pump trip causes a cold-shock and shortly after a hot-shock at the core outlet. This hot-shock corresponds to the onset of the natural circulation regime. The CFDxSys(meth1) evolution of Fig. 14 has been obtained from local temperature probes positioned inside the CFD domain similarly to the actual TC whereas the CFDxSys(meth2) evolution corresponds to the temperature at the core exit plane. The difference between these two former evolutions, which can be observed at the beginning of phase 2, exhibits that the core outlet temperature measurement is biased in the natural circulation regime. The observation of 3D temperature and velocity fields in the core outlet region have revealed that, in this regime, the sodium exiting the fuel sub-assemblies mixes with colder surrounding fluid before reaching the TC. The "Sys" evolution of Fig. 14, taken from the system code alone calculation, plots the temperature of the 0D element comprising the core outlet TC. This element collects the flows coming from the inner-core, the outer-core, the blanket and the reflector and shielding region. Hence, its initial temperature is close to the IHX primary inlet temperature (435 °C). Clearly, the numerical "Sys" evolution of the mean inner-core outlet temperature does not reproduce the experimental one during the onset of the natural circulation regime. However, the dynamics of the numerical prediction is correct during the second part of phase 2 and during phase 3.

5.3.5. The IHX primary inlet temperature

The evolution of the primary IHX inlet temperature is monitored with a sensor located in the upper part of the IHX inlet window. This measurement can be biased by thermal stratification in the natural circulation regime. Hence, for the Phénix NCT, a better estimation of the primary IHX inlet temperature is provided by the TC of the EIM pole located in the hot-collector at the elevation -500 corresponding to the middle of the IHX suction nozzle. Fig. 15 shows the experimental and CFD/system numerical evolutions of this temperature. The "Sys" evolution of Fig. 15, taken from the system code alone calculation, plots the temperature of the 0D element comprising the TC ML03-500. The CFD/system coupled model under estimates the ML03-500 temperature during phase 1 and the beginning of phase 2. This can be connected to the behavior already observed for the primary pump inlet and the core outlet temperatures. During the second part of phase 2, the increase of the ML03-500 temperature is over estimated by the CFD/system model. A sensibility calculation, which will not be described in detail in the present paper, has shown that the reflector and shielding region may play a significant thermal role during this phase. This region is comprised between the external radius of the blanket zone and the primary vessel, at the elevations of the fuel sub-assemblies (see Fig. 1). Due to heat exchanges through the primary vessel between the cold-collector and the aforementioned region, its average initial temperature is probably close to the pump inlet one. It is therefore a "cold" region. In the actual model, its mass and thermal exchanges with the hot-collector are accounted for in a very simplified manner. A more refined model is under-construction with the hope of reducing some of the experimental-numerical discrepancies observed in Fig. 15 during phase 2.

5.3.6. Hot-collector vertical temperature profile

The hot-collector vertical temperature profile is analyzed along the COLTEMP TC-pole that is located next to the UCS (see Fig. 8). In this paragraph, only the numerical predictions of the CATHARE/TRIO_U coupled calculation are presented.

The left part of Fig. 16 presents snapshots of the hot-collector vertical temperature profile at the beginning of phase 2. At t_1, a temperature peak is visible at an elevation corresponding to the bottom of the UCS. This peak results from the deflection of the core-outlet jet by the UCS. Then, the cold-shock stemming from the reactor scram is clearly visible at the elevation corresponding to the intersection between the core outlet jet and the COLTEMP TC-pole ($z = -1800$) whereas the rest of the collector remains at a constant temperature.

The right part of Fig. 16 presents the same snapshots as previously but focuses on phase 3. A general cooling rate of −30 to −35 °C/12600 s of the collector stemming from the opening of the steam generator air casing is clearly visible. A thermally stratified layer appears at the top of the hot-collector for times greater than $t_2 + 4200$ s. Prior to that, the upper-part of the collector is isothermal. Thus, the primary flow experiences a transition from mixing to stratification conditions during phase 3 that can be clearly seen in Video 2.

5.3.7. The primary mass flow-rate

Since the dedicated sensor is not reliable in natural circulation conditions, the experimental evolution of the primary mass flow-rate is not directly available from measurements for the Phénix NCT. However, it is possible to give an evaluation of the primary mass flow-rate using a thermal balance method. This method can be applied to the core or to the IHX. Since the core inlet/outlet temperature measurements are subject to considerable uncertainty, it was decided to apply it to the IHX. The method is the following. Knowing the secondary IHX mass flow-rate and the inlet/outlet temperatures, it is possible to deduce the primary/secondary heat transfer-rate. This quantity is used in an inverted thermal balance applied to the primary side of the IHX. Obtaining the primary mass flow-rate requires the primary inlet and outlet temperatures. This former parameter can be directly deduced from the measurements whereas the latter one is assumed to be equal to the secondary inlet temperature. This assumption is justified as long as the secondary mass flow-rate is greater than the primary one. The resulting experimental evolution is shown in Fig. 17 as well as the numerical predictions of the coupled CFD/system and of the system alone model.

It is interesting to notice that the experimental evolution is characterized by a drop occurring between $t_0 + 1500$ s and $t_0 + 2500$ s which is not well reproduced by the models. According to a sensitivity calculation, not detailed in the present paper, this phenomenon may stem from a secondary natural circulation loop occurring between the central fuel sub-assemblies and the external reflector and shielding region. The present model of the reflector and shielding region is probably too coarse to address this phenomenology. Again, a more refined model of this region is required to further improve the numerical results.

During the second part of phase 2, the IHX mass flow-rate is over-predicted both by the coupled CFD/system and the system alone models. This may derive from an over-prediction of the core inlet temperature by the models. During this phase, the diagrid is mainly fed with "hot" sodium flowing through the primary pumps and "cold" sodium brought by the flow reversal occurring in the main vessel cooling system. The resulting diagrid temperature is therefore very sensitive to the relative differences of motive and resistance momentum forces between these two parallel flow paths. Thus, the fact that the hydraulic resistance of the main vessel cooling system was chosen slightly higher in the coupled calculation may be responsible for the difference observed in the IHX mass flow-rate. During phase 3, the diagrid temperature is very similar to the pump outlet temperature. Hence, the better agreement between numerical and experimental evolutions observed during phase 3 for the coupled calculation (see Fig. 17) suggests that the pump outlet temperature is better predicted by the CFD/system model. This is consistent with the observation already made for the pump inlet temperature (see Fig. 11).

5.3.8. *Sodium free surface level*

Fig. 18 shows the experimental evolution of the free surface level of the main vessel cooling system. The experimental trends are well reproduced by the numerical models. Moreover, the "Sys" and "CFDxSys" evolutions are very similar. This validates the simplified model used in the present work to deal with the free surfaces of the hot- and cold-collectors.

5.4. *Sensitivity analysis for the CFD model of the hot-collector*

Additional calculations have been performed with the CFD model of the hot-collector in order to asses its sensitivity with respect to different sources of uncertainty in its inputs. These supplementary calculations have all been performed with the BC extracted from the same reference CATHARE/TRIO_U coupled calculation. Uncertain inputs are varied once at a time.

First, a calculation was performed by introducing a sub-grid turbulence model of the Smagorinsky type (with a Smagorinsky constant of 0.1) associated with standard wall functions. The primary IHX inlet temperature produced by this sensitivity calculation was found to be close from that of the reference coupled calculation. The same observation was made for the vertical temperature profiles along the ML03 and COLTEMP TC-poles. These may result from the fact that the turbulence intensity is low to moderate in the natural circulation regime and that the numerical schemes used for the present calculation promote numerical diffusion.

A second calculation relying on an isotropic and uniform grid refinement and a total number of meshes increased by a factor of 8 has been carried out. Though additional processors were used for this calculation, the corresponding computational time has risen up to 1.5 month. The flow paths in the hot collector (recirculation zones, direction of the core outlet jet after its deflection by the UCS) were found not to be significantly altered by the refinement of the grid at the initial steady state. Moreover, the primary IHX inlet temperature also remained close to that of the reference coupled calculation. Finally, the transition between mixing to stratified conditions observed at the beginning of phase 3 was found similar in both calculations. More generally, the vertical temperature profiles along the ML03 and COLTEMP TC-poles were not significantly modified by the mesh refinement.

The UCS is characterized by a complicated internal geometry immersed in sodium including the control rods, the instrumentation and grids. Thus, buoyancy driven internal sodium flows are likely to develop in the UCS so its "equivalent" thermal conductivity is probably higher than that of sodium. A third calculation was performed by increasing the "equivalent" thermal conductivity of the UCS by a factor of ten when compared to that of the reference coupled calculation. However, the results of this sensitivity calculation were found to be comparable to that of the reference calculation.

6. Summary and conclusion

The Phénix NCT has been addressed by coupling the system code CATHARE to the CFD code TRIO_U. The coupling methodology used is of the domain decomposition type associated to a defect correction approach. It allows for energy and momentum feedback from the CFD to the system scale. The level variations calculated by the CFD model are also accounted for at the system scale. The numerical methodology has been validated by comparing the results of a CATHARE/CATHARE coupled calculation to a CATHARE alone simulation. The CATHARE model used in the present paper derives from the data input deck previously developed in our group in order to validate the code for sodium applications (Pialla et al., 2011). Completely original detailed CFD models of the hot- and cold-collectors have been designed and realized for the present coupled CFD/system model.

The experimental vs. numerical comparisons performed against the available temperature measurements located at the inlets and outlets of the main components of the primary sodium system have illustrated the effectiveness of coupling CFD and system codes in addressing SFR transient. The results obtained by the system/CFD coupled model have demonstrated, for the first time, that several temperature measurements (e.g. core outlet temperature, IHX primary outlet temperature, etc.) are reliable in the forced convection regime but are biased in natural circulation conditions. These bias effects originate from local 3 dimensional effects that cannot be solved properly by a system code. As a consequence, the validation of a system code against these local temperature measurements cannot be straightforward. More generally, the time evolutions of the spatially averaged quantities calculated by a system code are smoother than the experimental one. If they are proved to be correct, the temperature fields obtained from the CFD/system coupled calculation in the CFD domains may eventually be used to define the spatially averaged quantities that can be directly compared to the results of a system code.

The system/CFD coupling has allowed obtaining an improved numerical vs. experimental agreement when compared to the system alone model. A part from the bias effects previously mentioned, this may result from the three dimensional nature of the primary flow affecting both collectors. Indeed, the coupled simulation has revealed the complexity and the diversity of the flow regimes during the Phénix NCT, both in forced and natural convection. This phenomenology seems too complex to address with a model based on a system code and relying on a basic 0D–1D grid.

According to the sensitivity analysis conducted for the present work, further improvements in the numerical vs. experimental agreement may firstly be obtained by decreasing the uncertainty associated with the IHX secondary inlet temperature. Secondly, it was found that the heat and mass exchanges between the hot-collector, the reflector and shielding region and the inter-wrapper region may play a significant role in the natural circulation regime yet there are not accounted for appropriately by the present model. A more realistic model is under construction to test this assumption. Finally, it was demonstrated that CFD mesh refinement or further improvement in the turbulence modeling is not expected to reduce some of the remaining experimental vs. numerical discrepancies.

Acknowledgements

The authors would like to acknowledge P. Gauthé, D. Tenchine and D. Pialla for answering our numerous questions about the operation of the Phénix reactor during the NCT. The authors are also grateful to the CEA and EDF teams involved in the Phénix end-of-life test campaign. Finally, assistance provided by B. Mathieu and G. Fauchet in the preparation of the snorky3D powered snapshots and videos was greatly appreciated.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at http://dx.doi.org/10.1016/j.nucengdes.2014.05.031.

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